

Kubernetes

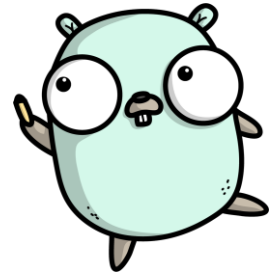


An Introduction

Before We Begin



Go here for install instructions



<https://goo.gl/1HV3Tu>

Kubernetes



An Introduction

\$ whoami - Bob



Bob Killen

rkillen@umich.edu

Senior Research Cloud Administrator

CNCF Ambassador

Github: [@mrbobbytables](https://github.com/mrbobbytables)

Twitter: [@mrbobbytables](https://twitter.com/mrbobbytables)



 **CLOUD NATIVE**
COMPUTING FOUNDATION

AMBASSADOR

\$ whoami - Jeff



Jeffrey Sica

jeef111x@gmail.com

Senior Software Engineer @  **Red Hat**

Github: [@jeefy](#)

Twitter: [@jeefy](#)



Cloud Native Computing Foundation (CNCF)



- Cloud Native Computing Foundation (CNCF) serves as the vendor-neutral home for many of the fastest-growing open source projects, including Kubernetes, Prometheus, and Envoy.
- <https://www.cncf.io/>



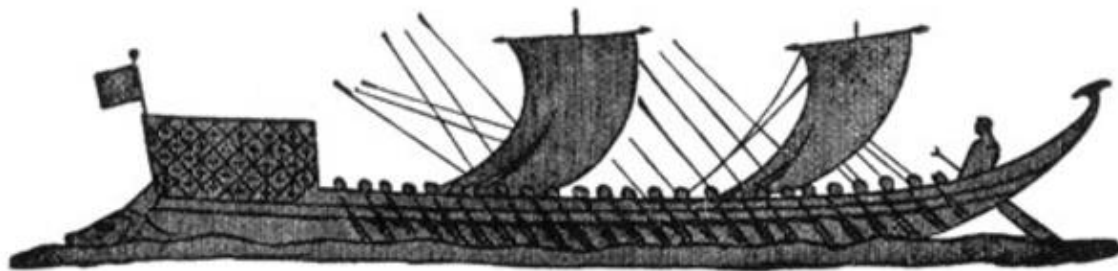
Project Overview

What Does “Kubernetes” Mean?



Greek for “pilot” or
“Helmsman of a ship”

키잡이



[Image Source](#)



What is Kubernetes?

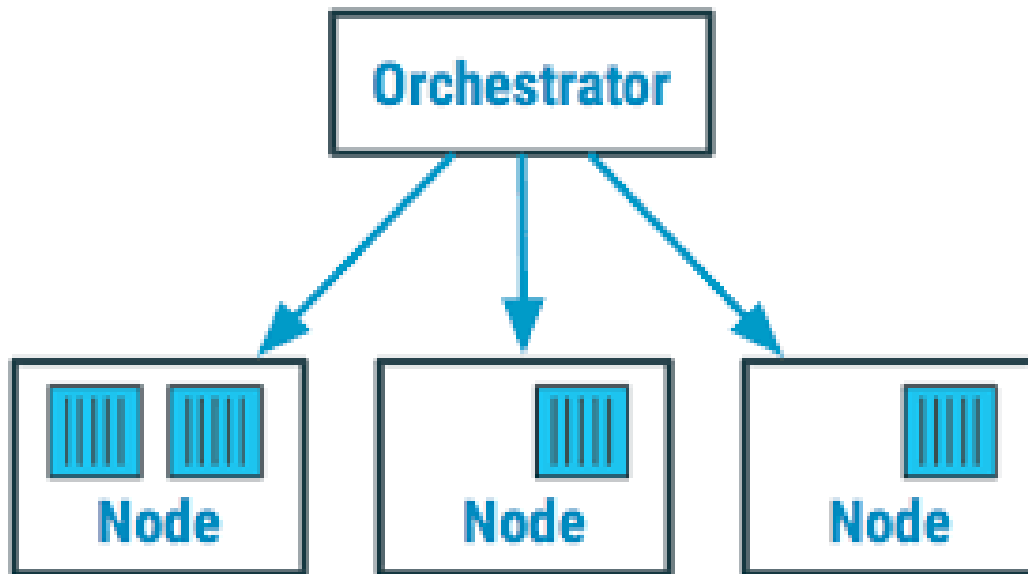


- Project that was spun out of Google as an open source **container orchestration platform**.
- Built from the lessons learned in the experiences of developing and running Google's Borg and Omega.

<https://dl.acm.org/doi/10.1145/2898442.2898444>

- Designed from the ground-up as a **loosely coupled** collection of components centered around **deploying, maintaining and scaling workloads**.

Container Orchestration



What is a Container Orchestration?



- Docker as a Container Platform
 - Container platforms like [Docker](#) are very popular these days to package applications based on a **microservices architecture**.
 - Containers can be made highly [scalable](#), which can be created [on-demand](#).
- Dynamic Container Management Problem on Demand
 - It becomes extremely difficult [to manage the container lifecycle](#) and its management when numbers [increase dynamically with demand](#).

What is a Container Orchestration?



- Container orchestration solves the problem by automating the scheduling, deployment, scalability, load balancing, availability, and networking of containers.
- Container orchestration is the automation and management of the lifecycle of containers and services.

14 Container Orchestration Tools for DevOps



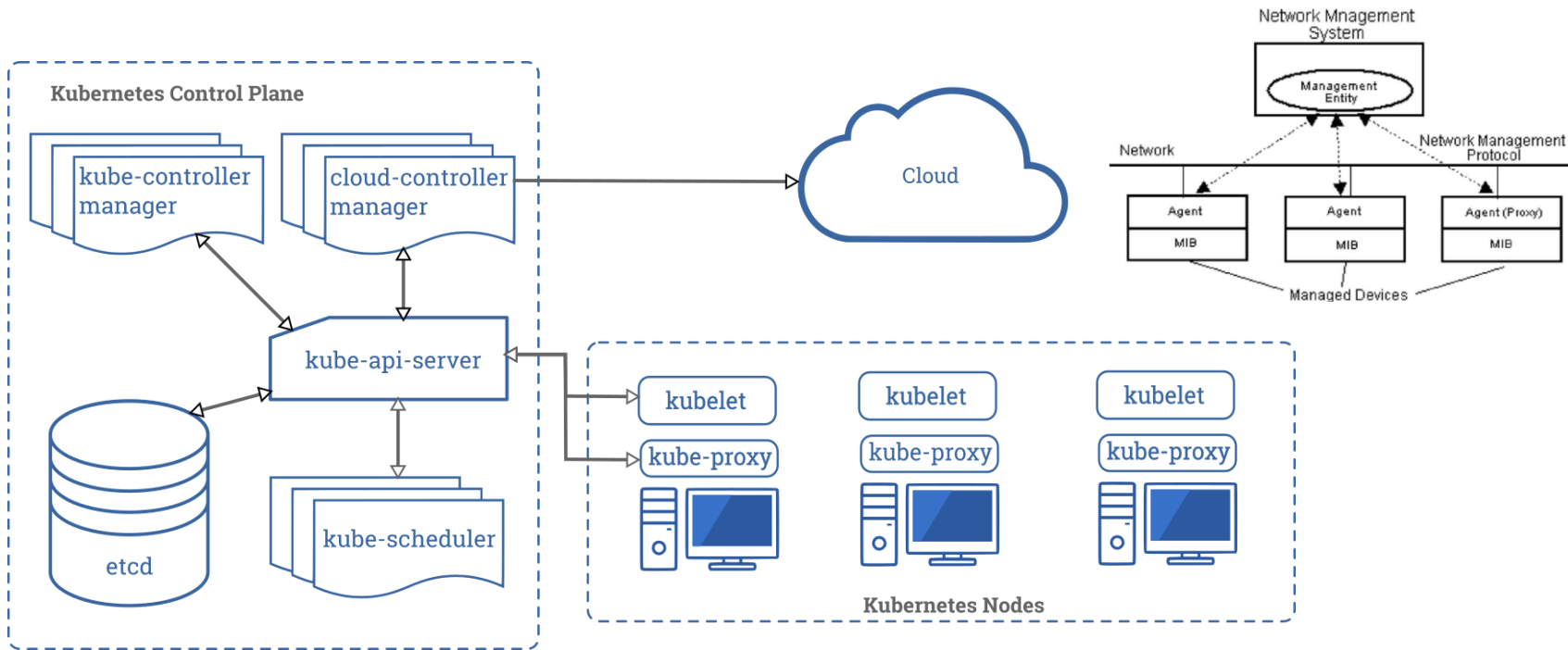
- [Kubernetes](#)
- [OpenShift](#)
- [Nomad](#)
- [Docker Swarm](#)
- [Docker Compose](#)
- [MiniKube](#)
- [Marathon](#)
- [Cloudfify](#)
- [Rancher](#)
- [Containership](#)
- [AZK](#)
- [GKE](#)
- [AKS](#)
- [AWS EKS](#)

Kubernetes



- Kubernetes is an open-source platform that was originally designed by Google and now maintained by the Cloud Native Computing Foundation.
- Kubernetes supports both declarative configuration and automation.
- It can help to automate deployment, scaling, and management of containerized workload and services.

Kubernetes Architecture





What Does Kubernetes do?

- Known as the **linux kernel of distributed systems**.
- **Abstracts away the underlying hardware** of the nodes and provides **a uniform interface** for workloads to be both deployed and consume **the shared pool of resources**.
- Works as an **engine** for resolving state by converging **actual** and the **desired state** of the system.

Decouples Infrastructure and Scaling



- **All services** within Kubernetes are natively **Load Balanced**.
- Can **scale up and down dynamically**.
- Used both to enable **self-healing** and **seamless upgrading or rollback** of applications.

Self Healing



Kubernetes will **ALWAYS** try and **steer the cluster to its desired state**.

- **Me:** “I want 3 healthy instances of redis to always be running.”
- **Kubernetes:** “Okay, I’ll ensure there are always 3 instances up and running.”
- **Kubernetes:** “Oh look, one has died. I’m going to attempt to spin up a new one.”

What can Kubernetes REALLY do?



- Autoscale Workloads
- Blue/Green Deployments (Blue:이전 버전, Green:새버전)
- Fire off jobs and scheduled cronjobs
- Manage Stateless and Stateful Applications
- Provide native methods of service discovery
- Easily integrate and support 3rd party apps

Most Importantly...



Use the **SAME API**
across bare metal and
EVERY cloud provider!!!



Who “Manages” Kubernetes?



**CLOUD NATIVE
COMPUTING FOUNDATION**

The CNCF is a child entity of the Linux Foundation and operates as a vendor neutral governance group.

Project Stats



- Over 55,000 stars on Github
- 2000+ Contributors to K8s Core
- Most discussed Repository by a large margin
- **70,000+** users in Slack Team



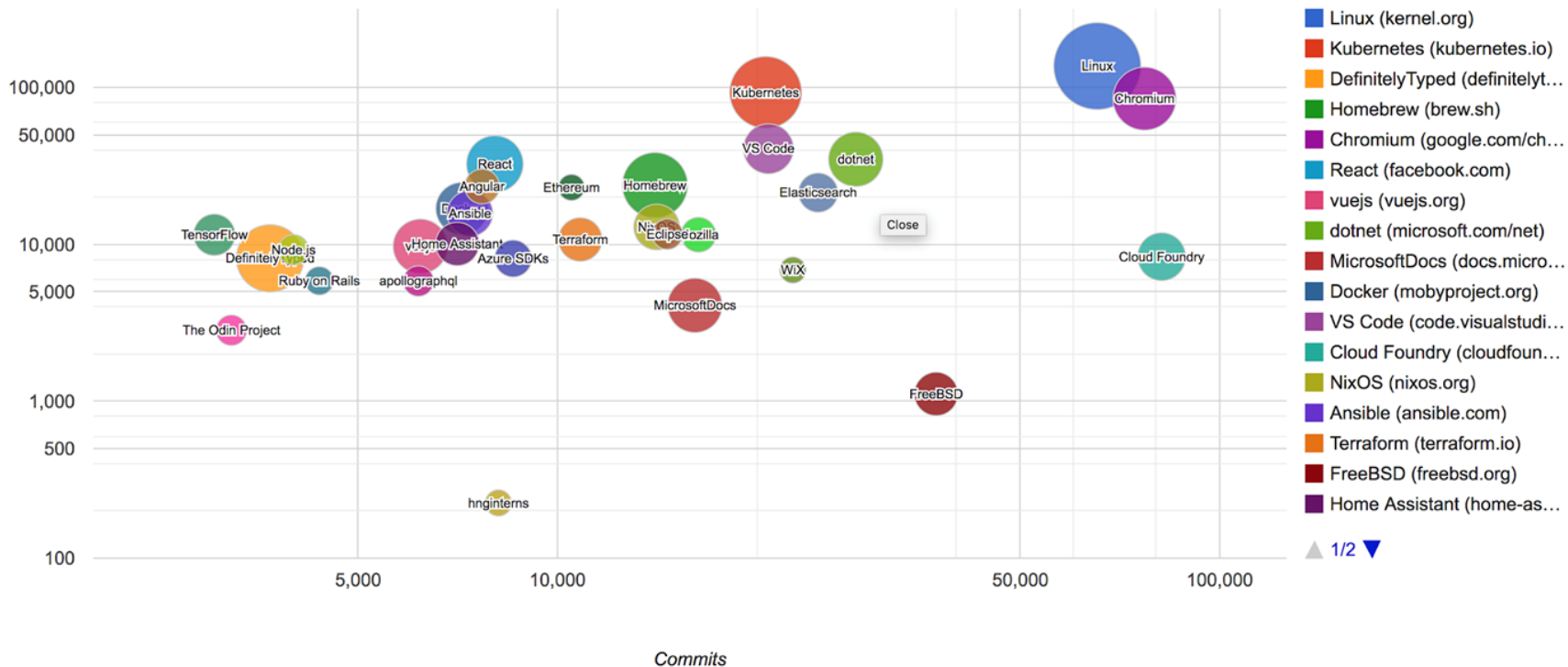
Project Stats

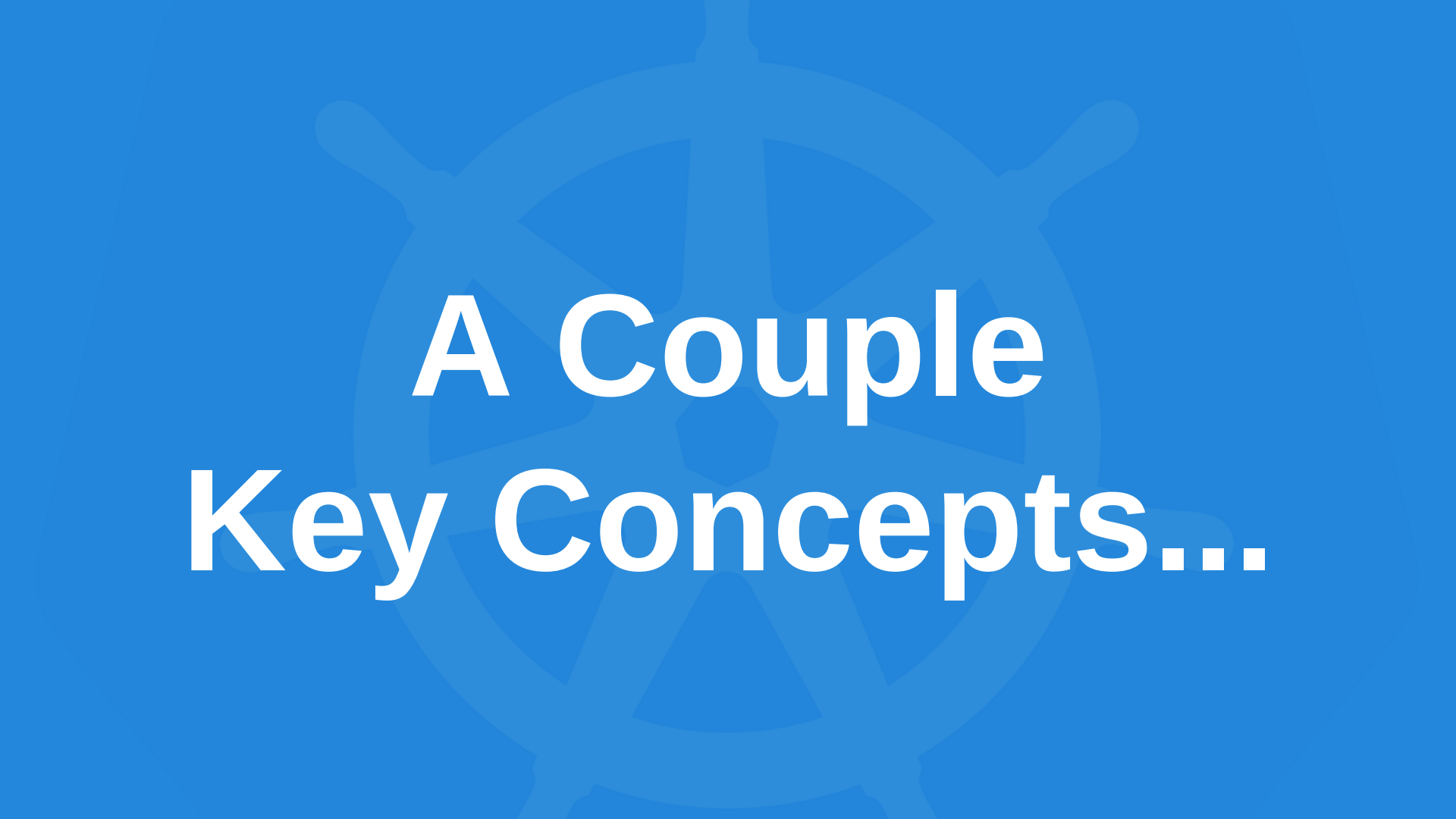


Top-30 projects in 2017

Pull
Requests
+ Issues

PRs + Issues



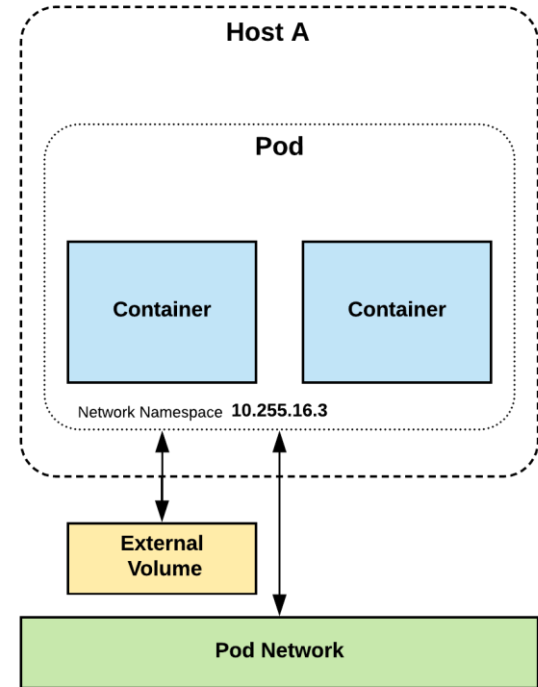


A Couple Key Concepts...

Pods



- **Atomic unit** or smallest “*unit of work*” of Kubernetes.
- Pods are **one or MORE containers** that share volumes, a network namespace, and are a part of a **single context**.

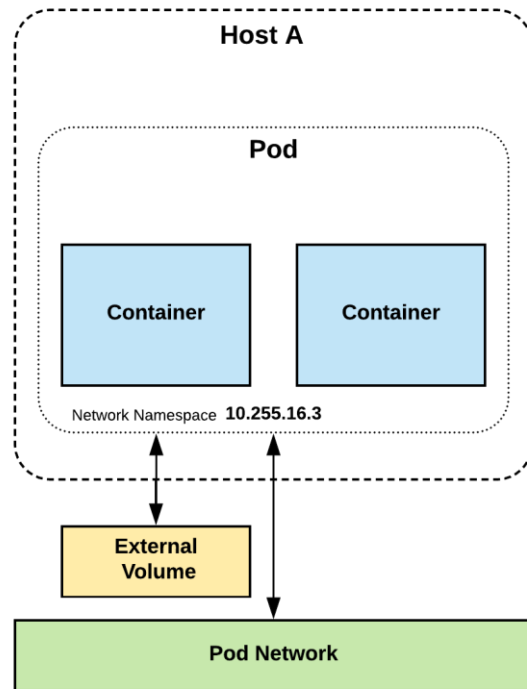


Pods



**They are
also
Ephemeral!**

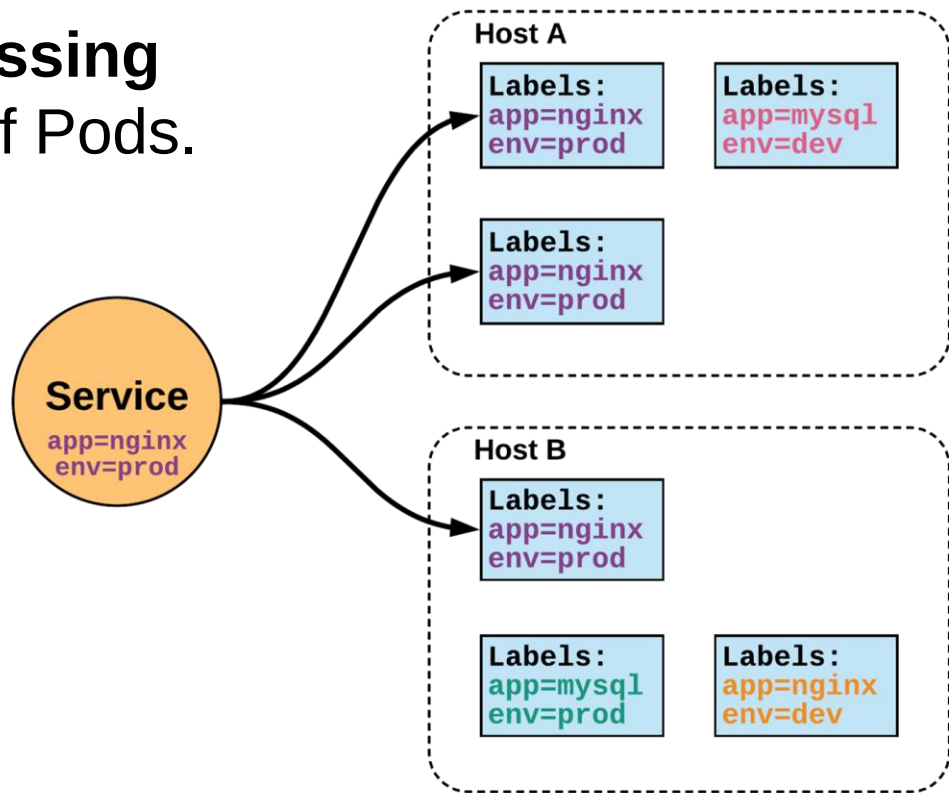
(일시적인)



Services



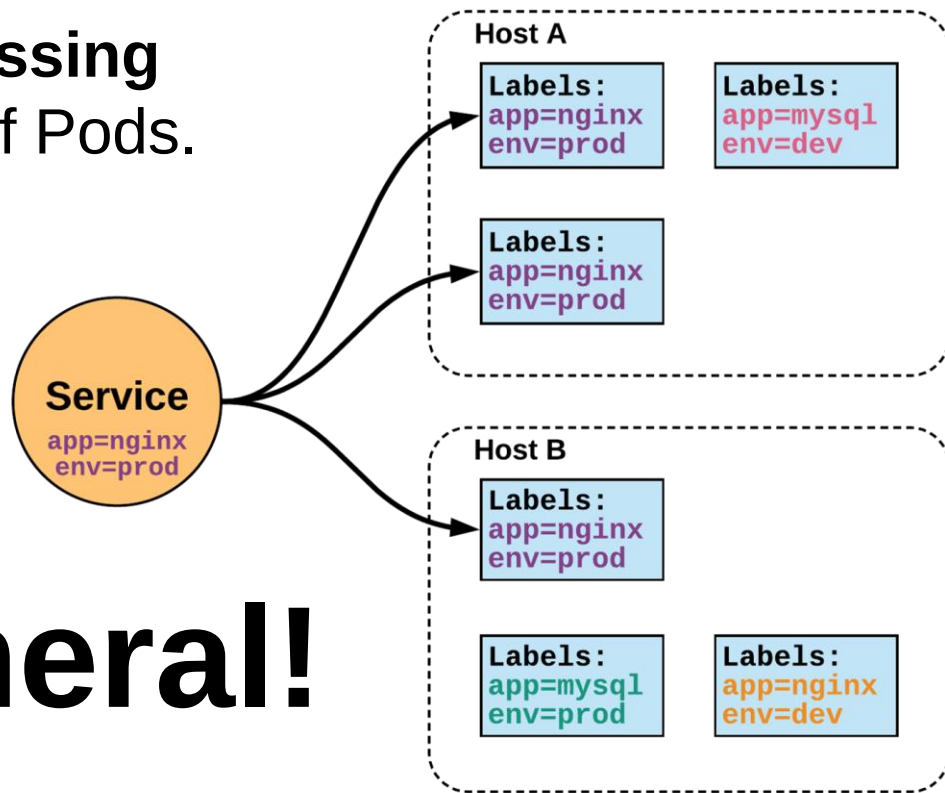
- **Unified method of accessing** the exposed workloads of Pods.
- **Durable resource**
(영속적인)
 - static cluster IP
 - static namespaced DNS name



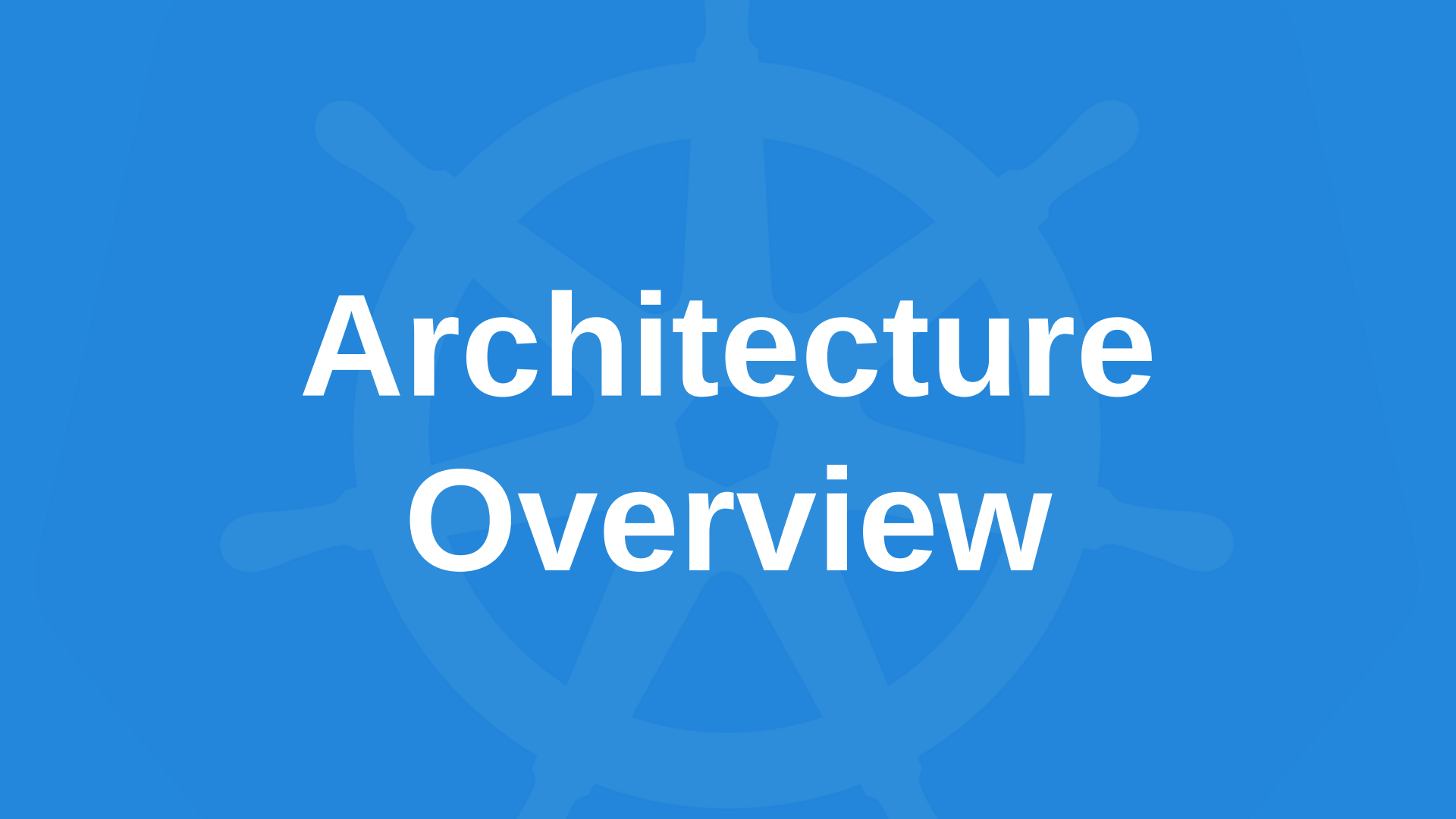
Services



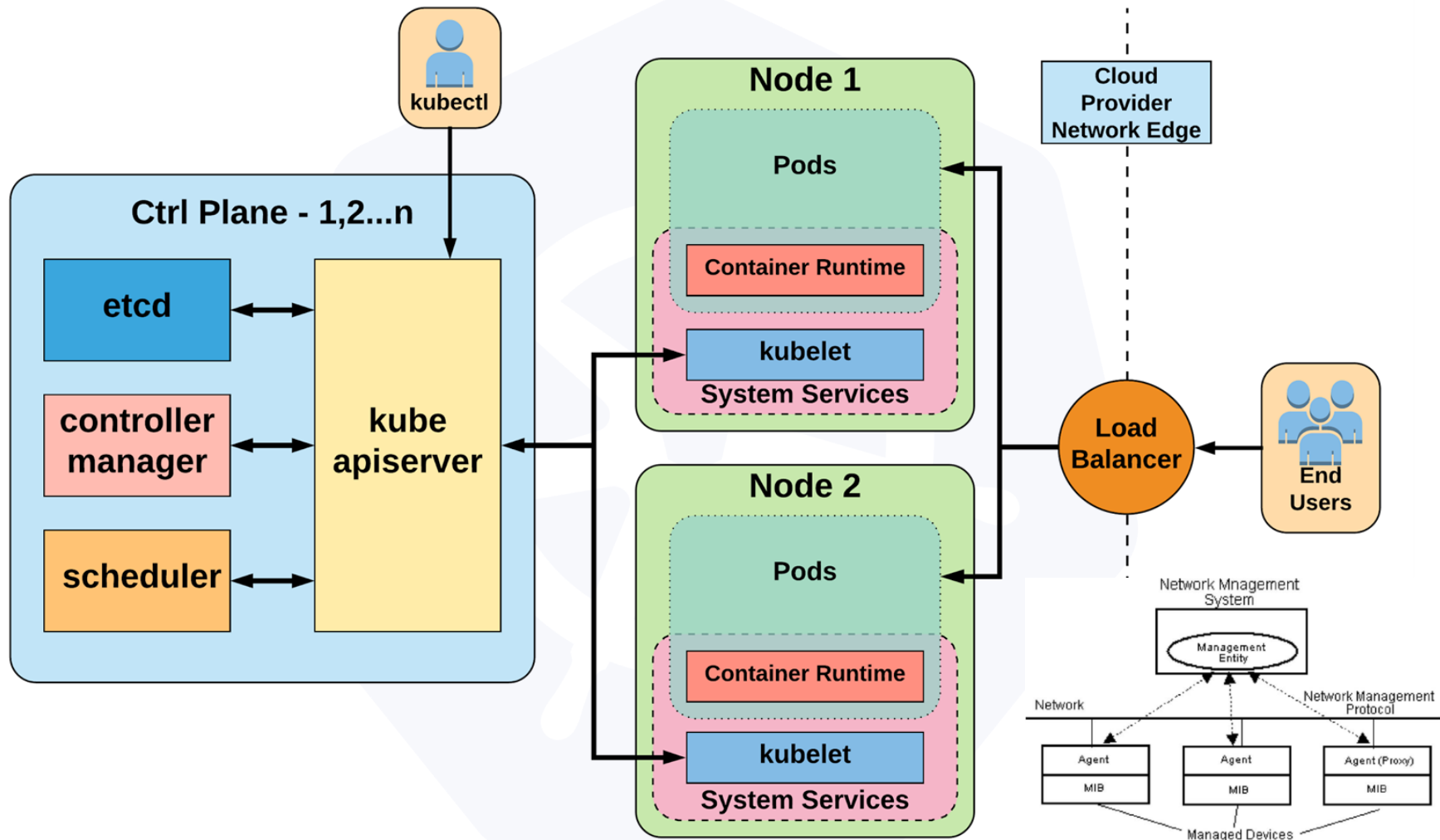
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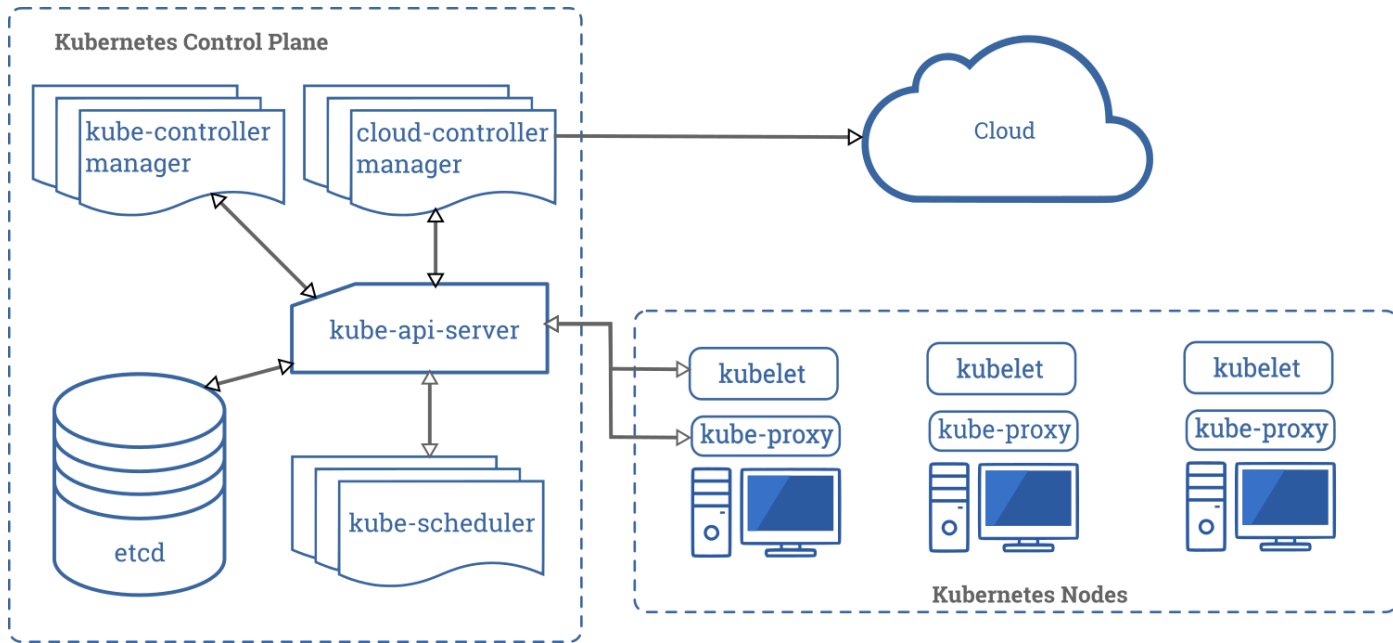
NOT Ephemeral!



Architecture Overview



Kubernetes Architecture





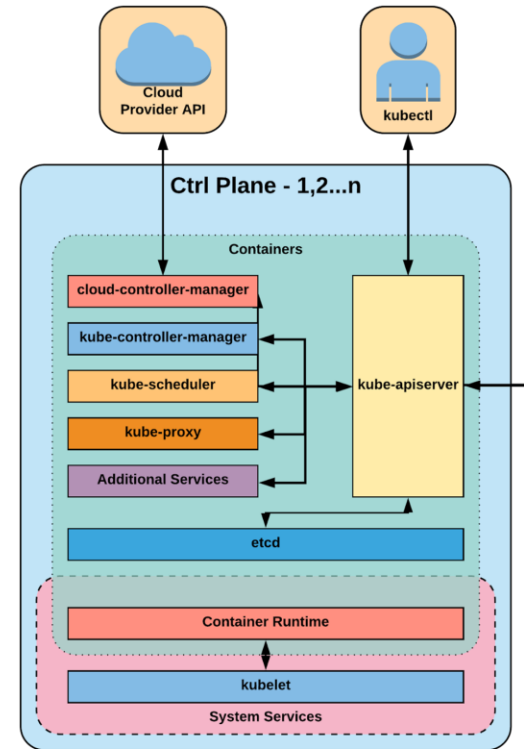
Control Plane Components

Architecture Overview

Control Plane Components



- kube-apiserver
- etcd
- kube-controller-manager
- kube-scheduler



kube-apiserver



- Provides a forward facing REST interface into the kubernetes control plane and datastore.
- All clients and other applications interact with kubernetes **strictly** through the API Server.
- Acts as the **gatekeeper** to the cluster by handling authentication and authorization, request validation, mutation, and admission control in addition to being the front-end to the backing datastore.

Etcd (<https://etcd.io/>)



- etcd acts as the **cluster datastore**.
- Purpose in relation to Kubernetes is to provide a strong, consistent and highly available **key-value store** for persisting cluster state.
- **Stores objects and config information.**



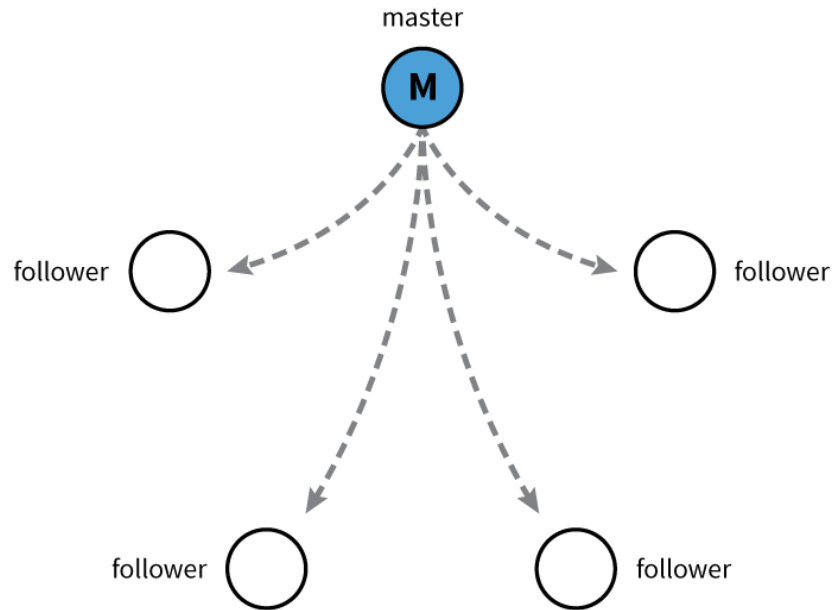
etcd



Uses “*Raft Consensus*” among a quorum of systems to create a fault-tolerant consistent “*view*” of the cluster.

<https://raft.github.io/>

<https://seongjin.me/raft-consensus-algorithm/>



[Image Source](#)



kube-controller-manager

- Serves as the **primary daemon** that manages all core component control loops.
- **Monitors the cluster state** via the apiserver and **steers the cluster towards the desired state.**

List of core controllers: <https://github.com/kubernetes/kubernetes/blob/master/cmd/kube-controller-manager/app/controllermanager.go#L344>

kube-scheduler



- Verbose **policy-rich engine** that evaluates workload requirements and attempts to place it on a matching resource.
- Default scheduler **uses bin packing**.
- **Workload Requirements** can include: general hardware requirements, affinity/anti-affinity, labels, and other various custom resource requirements.

The background of the slide features a large, faint, light blue Node.js logo, which is a ship's steering wheel, positioned on the right side.

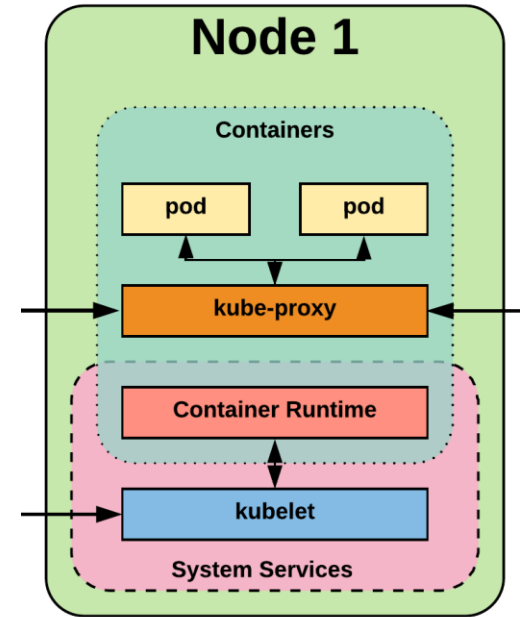
Node Components

Architecture Overview

Node Components



- kubelet
- kube-proxy
- Container Runtime Engine



kubelet



- Acts as the **node agent** responsible for **managing the lifecycle of every pod on its host**.
- Kubelet **understands YAML container manifests** that it can read from several sources:
 - file path
 - HTTP Endpoint
 - etcd watch acting on any changes
 - HTTP Server mode accepting container manifests over a simple API.

kube-proxy

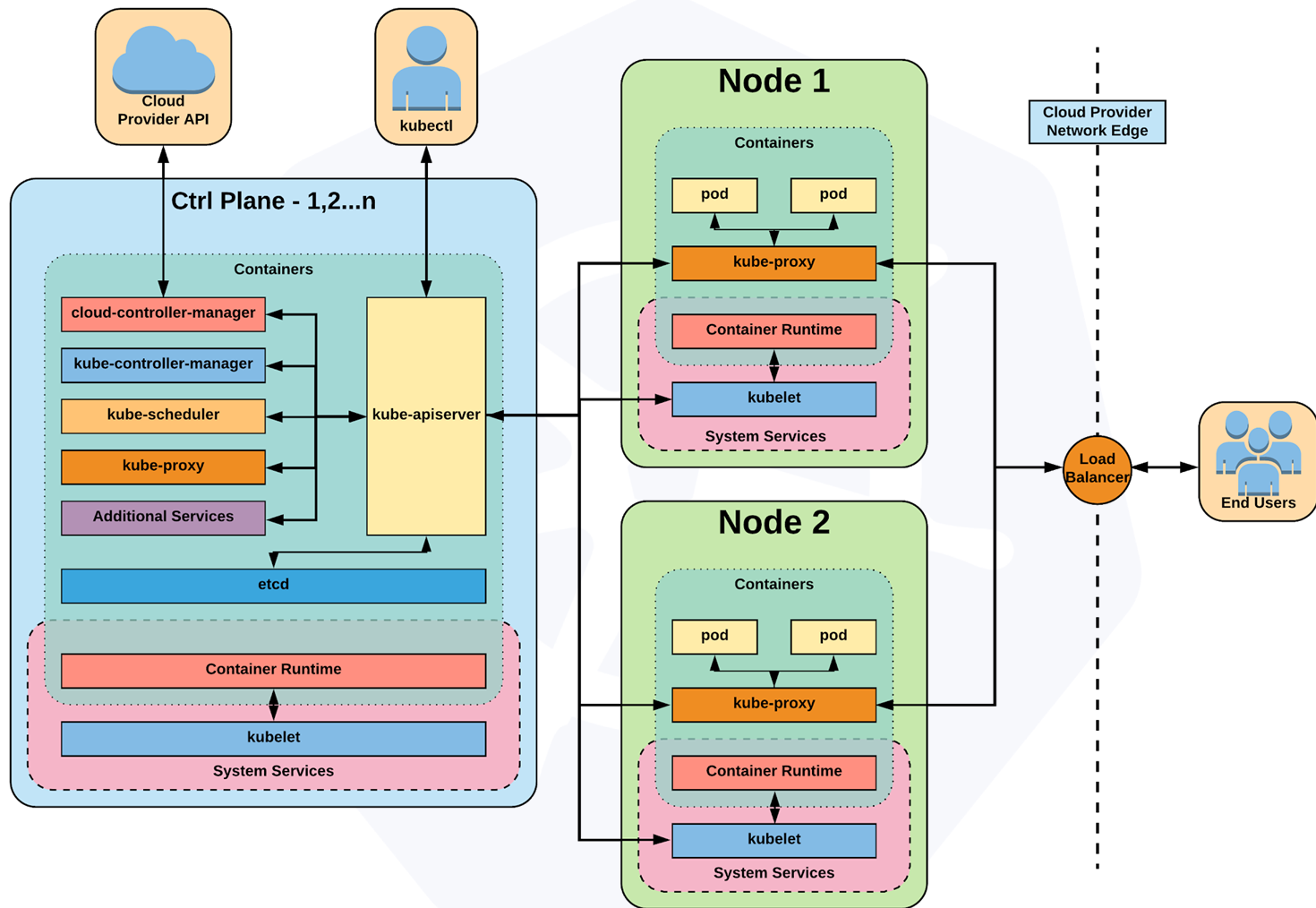


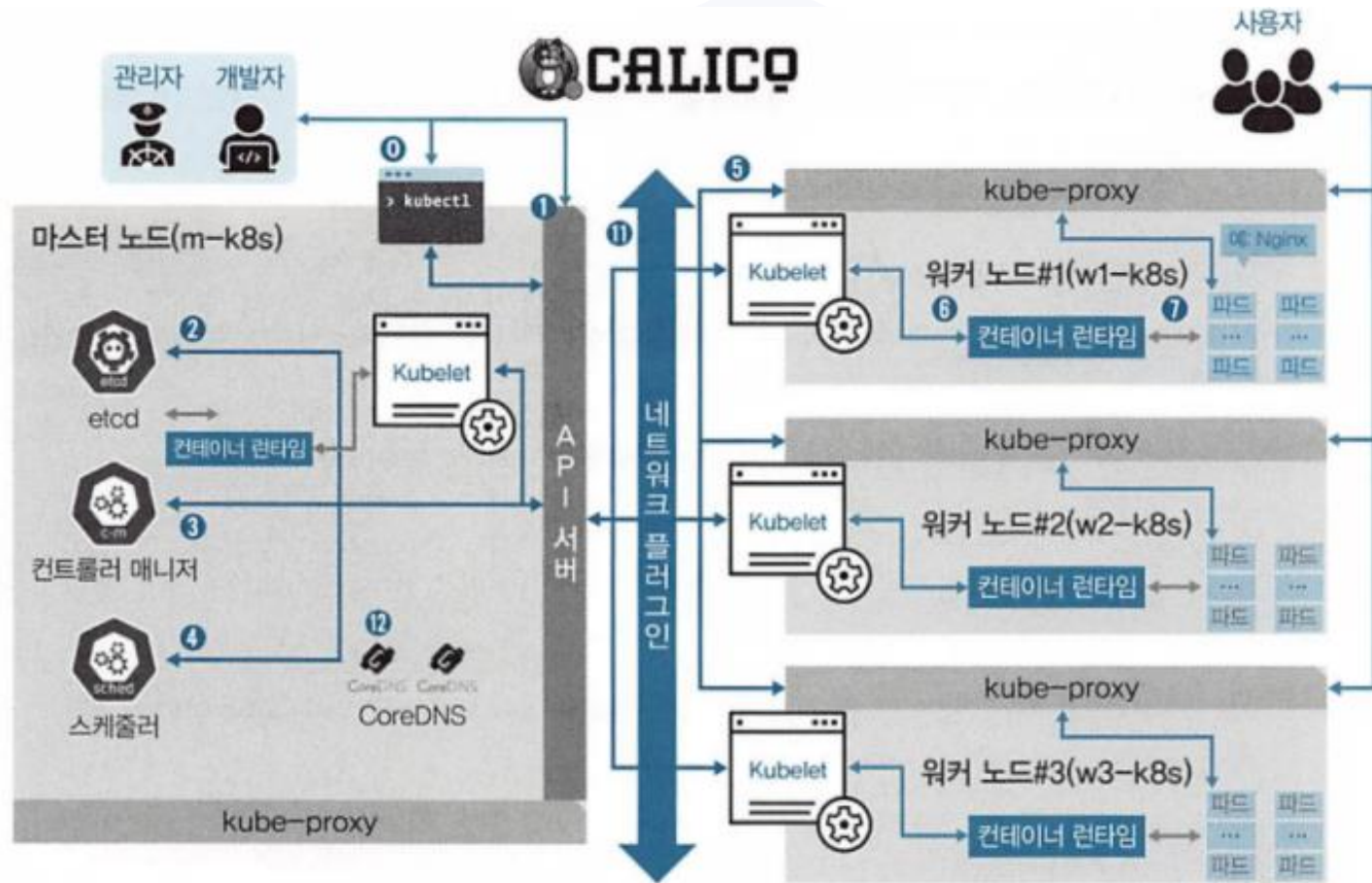
- Manages the network rules on each node.
- Performs connection forwarding or load balancing for Kubernetes cluster services.
- Available Proxy Modes:
 - Userspace
 - iptables
 - ipvs (default if supported)



Container Runtime Engine

- A container runtime is a **CRI (Container Runtime Interface)** compatible application that **executes and manages containers**.
 - Containerd (docker)
 - Cri-o
 - Rkt
 - Kata (formerly clear and hyper)
 - Virtlet (VM CRI compatible runtime)





(출처: 컨테이너 인프라 환경구축을 위한 쿠버네티스 도커, p96)



Optional Services

Architecture Overview

cloud-controller-manager



- Daemon that provides cloud-provider specific knowledge and integration capability into the core control loop of Kubernetes.
- The controllers include Node, Route, Service, and add an additional controller to handle things such as **PersistentVolume** Labels.

Cluster DNS

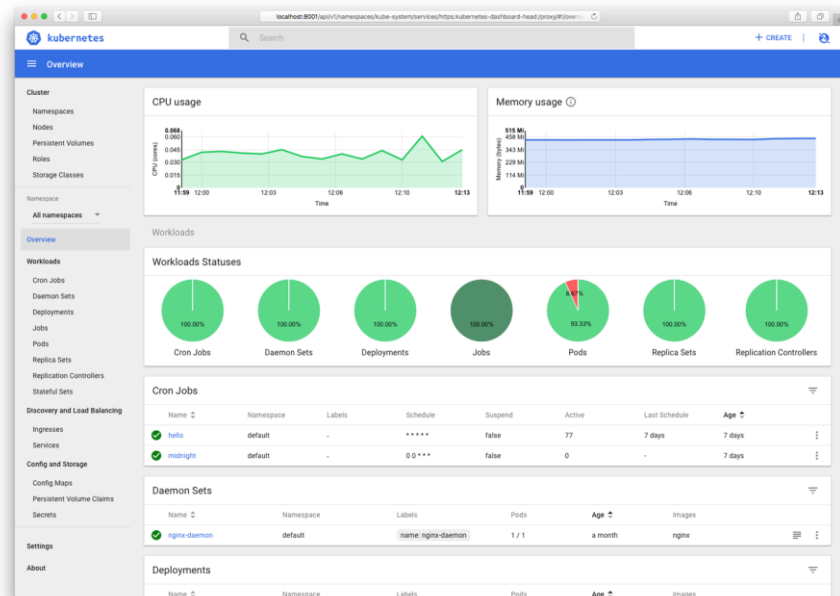


- Provides Cluster Wide DNS for Kubernetes Services.
 - Built on top of CoreDNS

Kube Dashboard



A limited, **general purpose web front end** for the Kubernetes Cluster.



Heapster / Metrics API Server



- Provides metrics for use with other Kubernetes Components.
 - Heapster (deprecated, removed last Dec)
 - Metrics API (current)



Networking

Architecture Overview

Kubernetes Networking



- **Pod Network**

- Cluster-wide network used for **pod-to-pod communication** managed by a CNI (Container Network Interface) plugin.

- **Service Network**

- Cluster-wide range of **Virtual IPs** managed by **kube-proxy** for service discovery.

Container Network Interface (CNI)

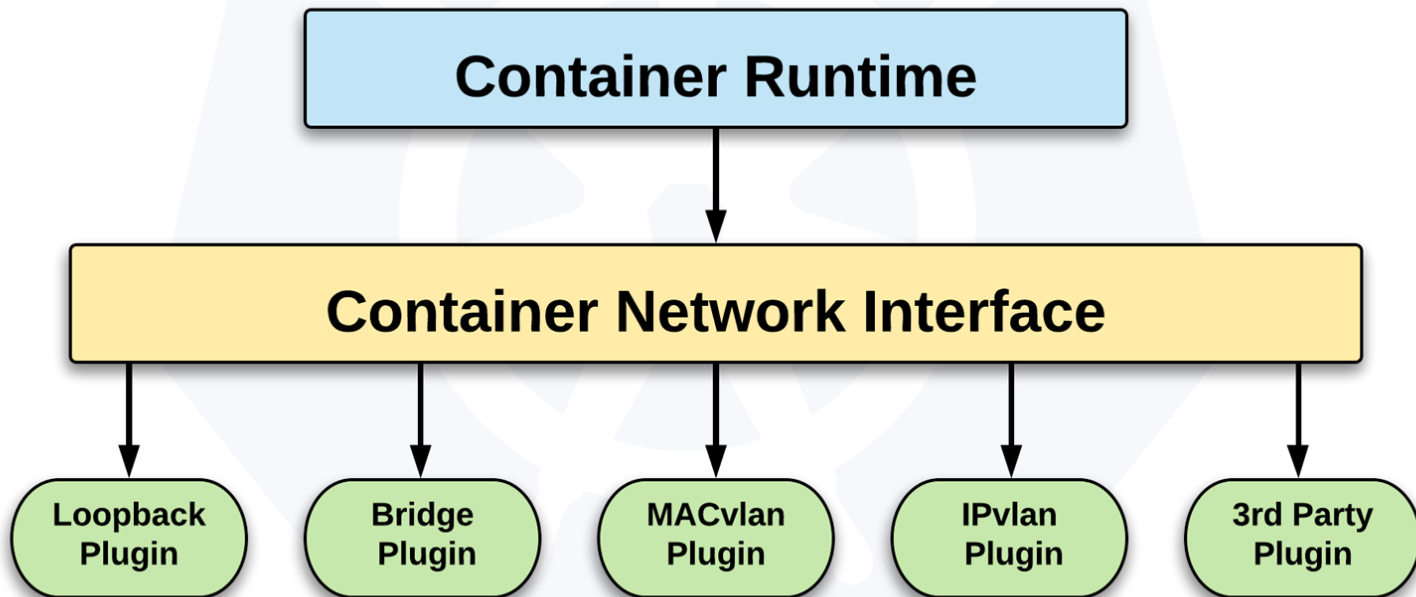


- Pod networking within Kubernetes is plumbed via the Container Network Interface (CNI).
- Functions as an interface between the container runtime and a **network implementation plugin**.
- CNCF Project
- Uses a simple JSON Schema.



C N I

CNI Overview



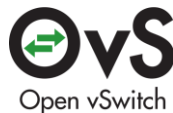
CNI Plugins



- Amazon ECS
- Calico
- Cilium
- Contiv
- Contrail
- Flannel



- GCE
- kube-router
- Multus
- OpenVSwitch
- Romana
- Weave



Fundamental Networking Rules



- All containers within a pod can communicate with each other unimpeded.
- All Pods can communicate with all other Pods without NAT.
- All nodes can communicate with all Pods (and vice-versa) without NAT.
- The IP that a Pod sees itself as is the same IP that others see it as.

Fundamentals Applied



- **Container-to-Container**

- Containers within a pod exist within the **same network namespace** and share an IP.
- Enables intrapod communication over *localhost*.

- **Pod-to-Pod**

- Allocated **cluster unique IP** for the duration of its life cycle.
- Pods themselves are fundamentally ephemeral.

Fundamentals Applied



- **Pod-to-Service**

- managed by **kube-proxy** and given a **persistent cluster unique IP**
- exists beyond a Pod's lifecycle.

- **External-to-Service**

- Handled by **kube-proxy**.
- Works in cooperation with a cloud provider or other external entity (load balancer).

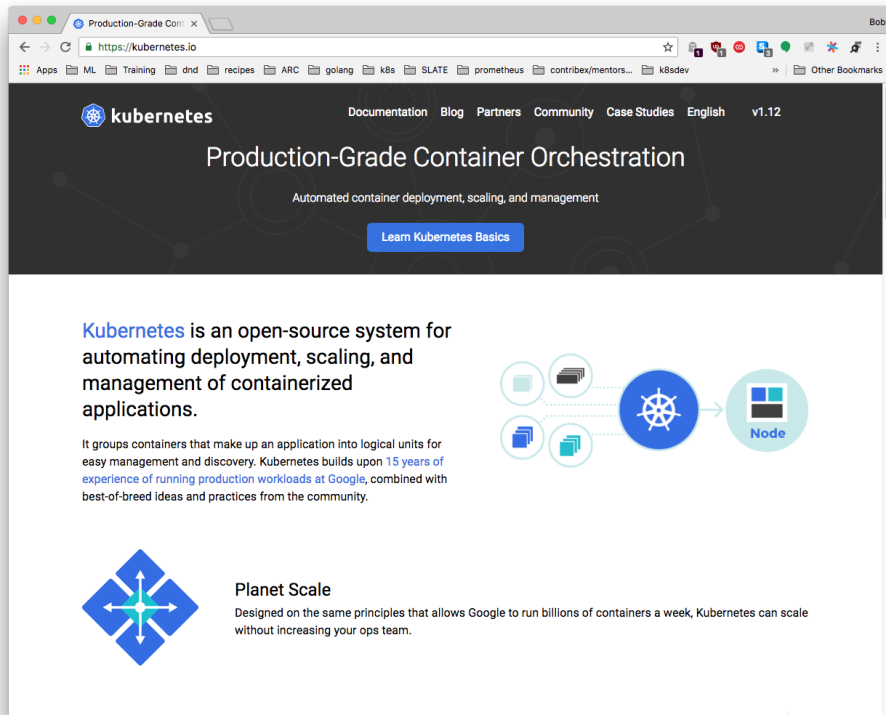
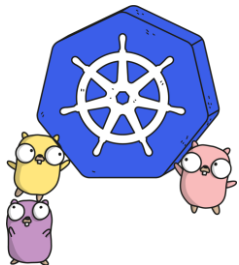


**Where to go
From Here**

kubernetes.io



Documentation
Examples
API Reference





Kubernetes

slack.k8s.io



70,000+ users
300+ public channels

CNCF

slack.cncf.io



8,200+ users
70+ public channels

Other Communities



- **Official Forum**

<https://discuss.kubernetes.io>

- **Subreddit**

<https://reddit.com/r/kubernetes>



- **StackOverflow**

<https://stackoverflow.com/questions/tagged/kubernetes>

Meetups



Kubernetes

meetup.com/topics/kubernetes/



850+ groups
380,000+ members

CNCF

meetups.cncf.io



200+ groups
120,000+ members
45+ countries

Conventions



KubeCon



CloudNativeCon

China:
TBD

North America:
November 18 - 21, 2019
San Diego, CA

Europe:
March 30 – April 2, 2020
Amsterdam, Netherlands

GitHub





Kubernetes

Kubernetes
<https://kubernetes.io> Verified

Repositories 65 **People** 664 **Teams** 305 **Projects** 2

Pinned repositories

kubernetes
Production-Grade Container Scheduling and Management
Go 42.2k 14.6k

features
Features tracking repo for Kubernetes releases
351 175

community
Kubernetes community content
2.6k 1.5k

website
Kubernetes website and documentation repo:
HTML 751 3.3k

test-infra
Test infrastructure for the Kubernetes project.
Go 637 573

examples
Kubernetes application example tutorials:
Shell 1.3k 809

Type: All **Language:** All

kube-state-metrics
Add-on agent to generate and expose cluster-level metrics.
Go 869 256 Apache-2.0 Updated 10 minutes ago

client-go
Go client for Kubernetes.
k8s-staging
Go 1,356 638 Apache-2.0 Updated 10 minutes ago

kubernetes
Production-Grade Container Scheduling and Management
containers kubernetes go cncf
Go 42,216 14,646 Apache-2.0 44 issues need help Updated 43 minutes ago

apiserver
Library for writing a Kubernetes-style API server.

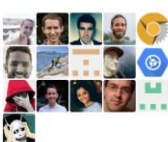
Top languages

Go HTML Shell Python JavaScript

Most used topics Manage

kubernetes k8s-staging containers cncf go

People 664 >



kubernetes / kubernetes [View Repository](#) [Watch](#) 2,747 [Unstar](#) 42,216 [Fork](#) 14,648

[Code](#) [Issues](#) 2,209 [Pull requests](#) 950 [Projects](#) 12 [Insights](#)

Filters **Labels** **Milestones** [New Issue](#)

2,209 Open ✓ 25,184 Closed

Author Labels Projects Milestones Assignee Sort

1 kubect! should accept --to-revision option **sig/api-machinery**
#69461 opened an hour ago by PierluigiLenociAkellus 1

1 Per service readiness probe **kind/feature** **needs-sig**
#69458 opened 4 hours ago by calind 1

1 pr:pull-kubernetes-typecheck flaked 20 times in the past week **needs-sig**
#69457 opened 6 hours ago by fejta-bot 1

1 Sample network policy is very unstable **kind/bug** **sig/network**
#69455 opened 7 hours ago by marcin-kasinski 1

1 kubect! diff segfaults when the object doesn't exist **kind/bug** **needs-sig**
#69450 opened 14 hours ago by smaslennikov 1

1 New field in Pod spec cause issues with DaemonSet **kind/bug** **sig/apps**
#69445 opened 16 hours ago by mortent 1

1 Failing/Flaking Test: E2E: [sig-autoscaling] [HPA] Horizontal pod autoscaling (scale resource: CPU) [sig-autoscaling] [Serial] [Slow] ReplicationController Should scale from 5 pods to 3 pods and from 3 to 1 and verify decision stability **kind/failing-test** **kind/flake** **priority/important-soon** **sig/autoscaling**
#69444 opened 16 hours ago by jberkus 1

3 AWS EBS volume resize is not idempotent **kind/bug** **sig/aws** **sig/storage**
#69434 opened 20 hours ago by leakingtapan 3

5 DUMMY issue / testing automation **kind/bug** **lifecycle/frozen** **priority/backlog** **sig/aws** **sig/cloud-provider** **sig/testing**
#69429 opened 23 hours ago by nikopen 5

1 Add an autoscaling step before cool off number for replicas in an hpa **kind/feature** **sig/autoscaling**
#69428 opened 23 hours ago by raravena80 1

SIGs



- Kubernetes components and features are broken down into smaller self-managed communities known as **Special Interest Groups (SIG)**.
- Hold weekly public recorded meetings and have their own mailing lists and slack channels.

SIG List

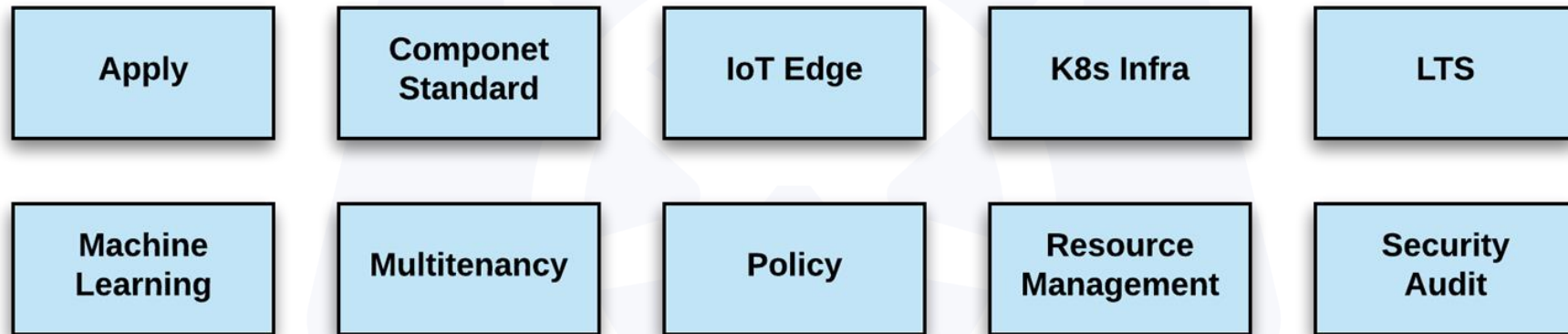
API Machinery	Apps	Architecture	Auth	Autoscaling	CLI
Cloud Provider	Cluster Lifecycle	Contributor Experience	Docs	Instrumentation	Multicluster
Network	Node	Product Management	Release	Scalability	Scheduling
Service Catalog	Storage	Testing	UI	Usability	Windows

Working Groups



- Similar to SIGs, but are topic focused, time-bounded, or act as a focal point for cross-sig coordination.
- Hold scheduled publicly recorded meetings in addition to having their own mailing lists and slack channels.

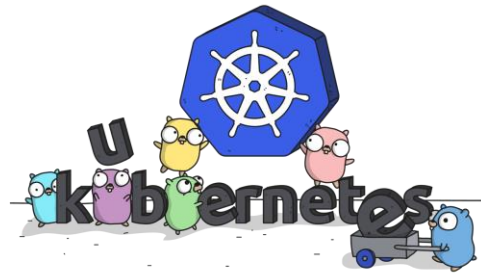
WG List



Links



- **Free Kubernetes Courses**
<https://www.edx.org/>
- **Interactive Kubernetes Tutorials**
<https://www.katacoda.com/courses/kubernetes>
- **Learn Kubernetes the Hard Way**
<https://github.com/kelseyhightower/kubernetes-the-hard-way>
- **Official Kubernetes Youtube Channel**
<https://www.youtube.com/c/KubernetesCommunity>
- **Official CNCF Youtube Channel**
<https://www.youtube.com/c/cloudnativefdn>
- **Track to becoming a CKA/CKAD (Certified Kubernetes Administrator/Application Developer)**
<https://www.cncf.io/certification/expert/>
- **Awesome Kubernetes**
<https://ramitsurana.gitbooks.io/awesome-kubernetes/content/>





Questions?